HemaTwin: Artificial Intelligence Based Molecular Digital Twin of Human Blood

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Abstract

Blood is a complex biological system composed of cellular elements, plasma proteins, metabolites, gases, and transient foreign agents such as pathogens. While clinical hematology characterizes blood primarily through cellular and biochemical assays, recent advances in structural biology and artificial intelligence enable molecular-level representation of blood components. In this work, we present a comprehensive framework for cataloging, storing, and comparing blood constituents using established biological databases and AlphaFold 3-based structural prediction [1]. By integrating proteomics, genomics, metabolomics, and pathogen sequence data with three-dimensional structural repositories such as the Protein Data Bank (PDB) [3], we propose a unified digital architecture for constructing a molecular digital twin of human blood. This framework facilitates standardized storage, structural comparison, and computational analysis of blood components for applications in diagnostics, systems biology, and drug discovery.

1 Introduction

Blood plays a central role in physiological homeostasis by transporting oxygen, nutrients, hormones, immune cells, and metabolic waste. Traditional blood analysis focuses on cellular composition and biochemical markers; however, the underlying molecular structures of blood proteins, nucleic acids, and metabolites govern their biological functions. Recent developments in high-throughput sequencing, mass spectrometry, and AI-based structure prediction provide an unprecedented opportunity to represent blood at atomic resolution as a molecular digital twin—a computational model capturing the three-dimensional structures and interactions of blood components. Next-generation sequencing (NGS) platforms enable comprehensive characterization of DNA and RNA from blood cells and cell-free nucleic acids, while mass spectrometry-based proteomics captures the protein landscape with high sensitivity, and liquid chromatography-mass spectrometry characterizes metabolites and small molecule ligands.

The Protein Data Bank (PDB) [3] serves as the primary repository of experimentally resolved macromolecular structures, while AlphaFold 3 [1] extends structure prediction to proteins, nucleic acids, and molecular complexes lacking experimental characterization. In parallel, databases such as GenBank [5], Ensembl [7], and the Human Metabolome Database (HMDB) [4] provide standardized sequence and metabolite references. Integrating these resources enables a systematic approach to constructing a cohesive digital twin of human blood, wherein individual molecular components are identified, structurally characterized, and assembled into functional complexes. This framework enables computational prediction of molecular interactions, drug binding, and disease-associated perturbations at atomic resolution.

2 Composition of Blood

Blood consists of both cellular and plasma components.

2.1 Cellular Components

Formed elements constitute approximately 45% of blood volume and include erythrocytes, leukocytes (neutrophils, lymphocytes, monocytes, eosinophils, basophils), and platelets. These components are typically identified through microscopy and flow cytometry and can be further characterized using transcriptomic and proteomic profiling.

2.2 Plasma Components

Plasma comprises water, proteins (albumin, globulins, fibrinogen), electrolytes, dissolved gases, nutrients, and metabolic waste products. Analytical methods such as liquid chromatography–mass spectrometry (LC-MS), ion chromatography, and blood gas analysis provide molecular-level identification.

2.3 Pathogens and Foreign Particles

Blood may contain transient foreign entities including viruses, bacteria, fungi, and parasites. Molecular detection methods include PCR, sequencing, and immunoassays.

3 Structural and Molecular Data Sources

3.1 Protein Structures: PDB and AlphaFold 3

Proteins constitute the dominant functional molecules in blood plasma and cells. When experimentally resolved structures exist, they are retrieved from the Protein Data Bank [3]. For proteins lacking structural data, AlphaFold 3 [1] enables high-accuracy prediction of three-dimensional conformations based on amino acid sequences.

3.2 Nucleic Acids: GenBank, Ensembl, and Structural Prediction

DNA and RNA sequences derived from blood cells and pathogens are stored and compared using genomic repositories such as GenBank [5] and Ensembl [7], enabling mutation analysis and phylogenetic comparison. Next-generation sequencing platforms [30, 31] generate raw sequence data that are processed through alignment [33, 34] or assembly pipelines to produce FASTA-formatted sequences. For structural representation, DNA and RNA sequences can be submitted to AlphaFold 3 [1] for prediction of three-dimensional nucleic acid structures or protein-nucleic acid complexes. Additionally, the Nucleic Acid Database provides experimentally determined structures for RNA and DNA molecules. The complete workflow from blood sample to nucleic acid structure is detailed in Section 6.

3.3 Metabolites and Ligands: HMDB and Structural Databases

Small molecules and metabolites are mapped against the Human Metabolome Database (HMDB) [4], which provides chemical structures, molecular weights, and pathway associations. Additional metabolite databases include METLIN, GNPS (Global Natural Products Social Molecular Networking), and MassBank, which offer mass spectral libraries for compound identification [46,47]. Following identification, metabolites are converted to chemical representations (SMILES, SDF formats) and three-dimensional structures are generated using cheminformatics tools [48, 49]. For protein-ligand interaction modeling, AlphaFold 3 [1] can predict three-dimensional structures of protein-metabolite complexes. The complete workflow from blood sample to ligand structure and protein-ligand complex prediction is detailed in Section 7.

3.4 Pathogens: Viral and Microbial Databases

Sequenced pathogens are compared against virus and microbial genome databases for identification and strain classification.

4 Framework for Storage and Comparison

We propose a modular framework in which each blood component is represented according to its molecular type. Table 1 summarizes the database mapping and data representation for each component type.

Structural comparison between predicted and experimental protein models can be performed using root-mean-square deviation (RMSD) and interface analysis. Confidence metrics from AlphaFold predictions guide quality control.

Table 1: Framework for blood component storage and comparison

Component	Database	Representation
Proteins	PDB / AlphaFold 3	3D coordinates
DNA/RNA	GenBank / Ensembl / AlphaFold 3	Sequences & 3D structures
Metabolites	HMDB / METLIN / AlphaFold 3	Chemical structures & complexes
Pathogens	Virus databases	Genome sequences

5 Detailed Implementation: Protein Identification Workflow

Proteins constitute the most structurally complex components of blood, requiring a sophisticated multi-step pipeline from biological sample to three-dimensional structure. This section describes the complete workflow for converting raw blood samples to molecular-level protein representations stored in PDB format.

5.1 Mass Spectrometry Data Acquisition

Blood proteins are analyzed using high-resolution mass spectrometry instruments such as the Thermo Scientific Orbitrap Fusion Lumos Tribrid, Orbitrap Elite Hybrid Ion Trap-Orbitrap, and Q Exactive Hybrid Quadrupole-Orbitrap [22]. These instruments produce proprietary binary files in RAW format containing mass-to-charge (m/z) spectra and intensity data. The RAW format captures comprehensive information about peptide fragmentation patterns essential for subsequent identification.

5.2 Conversion from RAW to mzML Format

The first computational step involves converting proprietary RAW files to the open mzML (Mass Spectrometry Markup Language) format using ProteoWizard's msconvert tool [8]. The mzML format [9] is a community standard established by the Human Proteome Organization Proteomics Standards Initiative (HUPO-PSI), ensuring vendor neutrality and compatibility with downstream analysis tools. This conversion maintains high metadata fidelity while enabling interoperability across software platforms. The msconvert tool applies peak-picking algorithms to reduce file size while preserving spectral quality, with minimal information loss when properly configured.

5.3 Peptide Identification from MS/MS Spectra

Peptide identification is performed using database search engines such as MSFragger [10], MaxQuant [12], or Comet [15]. These tools match experimental tandem mass spectrometry (MS/MS) spectra against theoretical spectra generated from known protein sequence databases (e.g., UniProt [6]). The search process yields peptide-spectrum matches (PSMs) with associated confidence scores. Modern high-resolution instruments achieve identification rates of 60–90% depending on sample complexity and database coverage. False discovery rate (FDR) thresholds, typically set at 1%, ensure statistical confidence in peptide identifications [21].

5.4 Protein Inference and Accession Mapping

Identified peptides are aggregated into protein groups using tools such as ProteinProphet [20] or Philosopher [14]. This protein inference step resolves ambiguities arising from shared peptides between protein isoforms. Each identified protein is mapped to database accession identifiers from UniProt [6], enabling retrieval of full-length amino acid sequences. The success rate for mapping confidently identified peptides to known protein accessions exceeds 95% for well-annotated organisms such as human.

5.5 FASTA Sequence Extraction

Protein accession identifiers are used to retrieve complete amino acid sequences in FASTA format from UniProt [6] or similar repositories. The FASTA format provides a standardized text representation of protein sequences, serving as the input for structure prediction algorithms.

5.6 Three-Dimensional Structure Prediction

FASTA sequences are submitted to AlphaFold 3 [1] or similar AI-based structure prediction platforms [2] to generate three-dimensional atomic coordinates. AlphaFold 3 produces PDB-formatted files containing predicted x, y, z coordinates for all atoms in

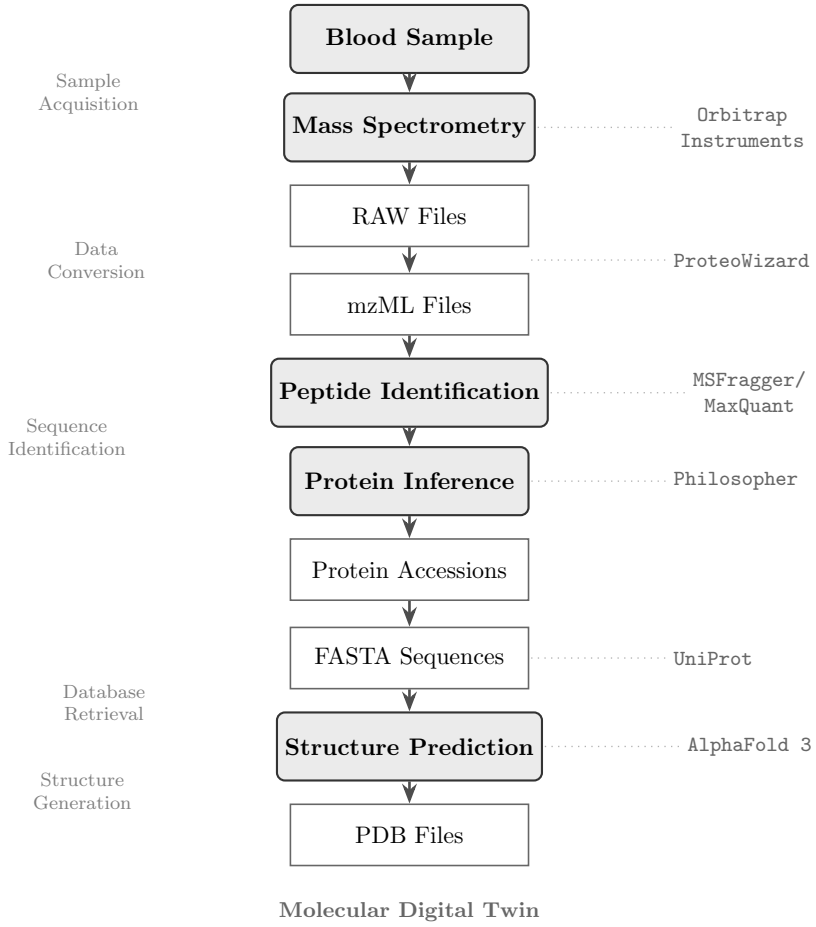

Figure 1: End-to-end workflow for protein identification and structural representation in blood digital twin construction. The pipeline transforms raw blood samples into molecular-level 3D structures through sequential stages of mass spectrometry analysis, data conversion, peptide identification, protein inference, and AI-based structure prediction. Gray boxes indicate processing steps, white boxes represent data formats, and monospace text shows the primary computational tools employed at each stage.

the protein structure, along with confidence metrics (pLDDT scores) indicating the reliability of each residue's predicted position. For proteins with experimentally determined structures available in the Protein Data Bank [3] or the AlphaFold Protein Structure Database [25], experimental or pre-computed coordinates are retrieved directly, bypassing prediction.

5.7 Fallback Strategies for Unidentified Proteins

When standard database searches fail to identify peptides, several fallback strategies can be employed. De novo sequencing tools (PEAKS [16], DeepNovo [17], Novor [18]) predict peptide sequences directly from spectra without reliance on databases, achieving 85–95% accuracy for short peptides. Open modification searches using MSFragger [11] or MetaMorpheus [19] detect unexpected post-translational modifications or mass shifts. Proteogenomics [24] approaches integrate patient-specific RNA-seq data to construct custom databases containing novel splice variants or somatic mutations, particularly valuable in precision oncology applications.

5.8 End-to-End Workflow Summary

The complete pipeline from blood sample to structural representation is illustrated in Figure 1. This comprehensive workflow integrates established proteomics methods with AI-based structure prediction to generate molecular-level representations of blood proteins. The pipeline begins with biological sample acquisition and mass spectrometry analysis, proceeds through bioinformatics processing and database matching, and culminates in three-dimensional structural models stored in PDB format. This workflow has been validated in clinical proteomics facilities including the Clinical Proteomic Tumor Analysis Consortium (CPTAC) [23], Broad Institute, and major research hospitals.

6 Nucleic Acid Identification Workflow

Nucleic acids (DNA and RNA) represent the genetic and transcriptomic components of blood, requiring next-generation sequencing-based approaches for molecular characterization. This section describes the complete workflow for converting blood samples to nucleic acid sequences and three-dimensional structural representations.

6.1 Biological Source and Sample Preparation

DNA in blood is primarily found in white blood cells (WBCs), as mature erythrocytes lack nuclei. Cell-free DNA (cfDNA) can be isolated from plasma for liquid biopsy applications [29]. RNA is present in WBCs and plasma as messenger RNA (mRNA), microRNA (miRNA), and cell-free RNA (cfRNA). Whole blood, plasma, or isolated cell fractions serve as starting material depending on the analytical objective.

6.2 Nucleic Acid Extraction

Nucleic acid extraction isolates pure DNA or RNA molecules from biological samples using silica column-based kits (Qiagen, Thermo Fisher Scientific) or magnetic bead-based automated platforms (KingFisher). Extraction protocols are optimized for yield and purity, with quality assessed by spectrophotometry (260/280 nm ratio) and integrity verified through electrophoresis or bioanalyzer systems [30]. Clinical-grade extraction achieves greater than 99% recovery efficiency for high-quality samples. For RNA extraction, RNase inhibitors and rapid processing prevent degradation.

6.3 Library Preparation

Extracted nucleic acids undergo library preparation to generate sequencing-compatible molecules. For DNA libraries, genomic DNA is fragmented (mechanically or enzymatically), followed by end-repair, adapter ligation, and PCR amplification to add sequencing primers and sample barcodes. RNA libraries require reverse transcription to complementary DNA (cDNA), with optional ribosomal RNA depletion or poly-A selection to enrich for mRNA. Adapter ligation and amplification complete library construction [32]. Library quality is assessed through fragment size distribution analysis and quantification via fluorometry or quantitative PCR.

6.4 Next-Generation Sequencing

Prepared libraries are sequenced on high-throughput NGS platforms. Illumina systems (NovaSeq, NextSeq) dominate clinical applications due to high accuracy (error rate approximately 0.1%), scalability, and established regulatory acceptance [30, 31]. Alternative platforms include Pacific Biosciences (PacBio) HiFi sequencing for long reads and Oxford Nanopore for real-time sequencing. Sequencing generates FASTQ files containing raw sequence reads

with per-base quality scores (Phred scores) encoding confidence in base calls.

6.5 Quality Control and Read Processing

Raw FASTQ files undergo quality control using FastQC [36] to assess read quality metrics, sequence duplication, adapter contamination, and GC content distribution. Low-quality bases and adapter sequences are trimmed using Trimmomatic or Cutadapt. Reads failing quality thresholds are filtered to ensure downstream analysis accuracy. This preprocessing step is critical for clinical applications where false positives can impact diagnostic decisions.

6.6 Sequence Alignment and Assembly

Processed reads follow one of two analytical paths depending on application. For clinical genomics with available reference genomes, reads are aligned to reference sequences using BWA [33] for DNA or STAR [34] for RNA. Alignment generates SAM/BAM files containing read positions and mapping quality scores, processed using SAMtools [35]. For de novo applications without reference genomes (e.g., pathogen discovery), reads are assembled into contiguous sequences (contigs) using SPAdes for DNA or Trinity [37] for RNA. Assembly quality is evaluated through N50 metrics and coverage statistics.

6.7 FASTA Sequence Generation

For RNA sequencing data, transcript reconstruction and isoform quantification are performed using tools such as StringTie [41], Cufflinks [42], or RSEM [43] to identify alternative splicing events and generate multiple transcript isoforms per gene. RNA splicing—the process of removing introns and joining exons in different combinations—produces multiple mature mRNA variants from a single gene locus. Each splice variant encodes a distinct protein isoform with potentially different functional properties and three-dimensional structures.

Consensus sequences are extracted from aligned or assembled data and formatted as FASTA files—the universal text-based representation of nucleotide sequences. For reference-aligned RNA data, transcript assembly generates multiple FASTA sequences per gene corresponding to detected splice variants. For DNA sequencing, consensus calling uses bcftools or GATK [38] to incorporate variant information. For assembled data, contig sequences are directly exported. FASTA format consists of a header line (beginning with "¿") containing sequence identifiers followed by nucleotide sequences (A, T, G, C for DNA; A, U, G, C for RNA). Quality metrics include coverage depth, isoform expression levels, and base-call confidence.

6.8 Three-Dimensional Structure Prediction

FASTA-formatted DNA and RNA sequences serve as input for three-dimensional structure prediction using AlphaFold 3 [1], which can model nucleic acid structures and protein-nucleic acid complexes. For RNA sequencing data where alternative splicing has generated multiple transcript isoforms, each isoform is translated to its corresponding protein sequence and modeled independently, producing distinct three-dimensional structures for each splice variant. This is critical as alternative splicing can alter protein domains, binding sites, and overall structure—for example, the human DSCAM gene produces over 38,000 splice variants, each potentially with unique structural features [44].

Input sequences are formatted in AlphaFold 3's JSON specification, designating nucleotide chains for RNA structures or amino acid sequences for proteins derived from spliced transcripts. The prediction generates mmCIF structure files containing atomic coordinates and confidence metrics (pLDDT scores) for each residue. When multiple isoforms are present, separate structure predictions are performed for each variant, creating a library of structures representing the full diversity of gene products. Experimentally determined RNA and DNA structures from the Nucleic Acid Database or PDB [3] can supplement predicted models where available.

6.9 Fallback Strategies for Complex Variants

When standard short-read sequencing fails to resolve structural variants, repeat regions, or full-length transcripts, long-read sequencing platforms (PacBio HiFi, Oxford Nanopore) provide kilobase-scale contiguous reads [39]. Targeted sequencing via PCR amplification or hybrid capture enriches specific genomic regions for deep coverage. Metagenomic sequencing identifies unknown pathogens or foreign nucleic acids using tools such as Kraken2 or MetaSPAdes [40].

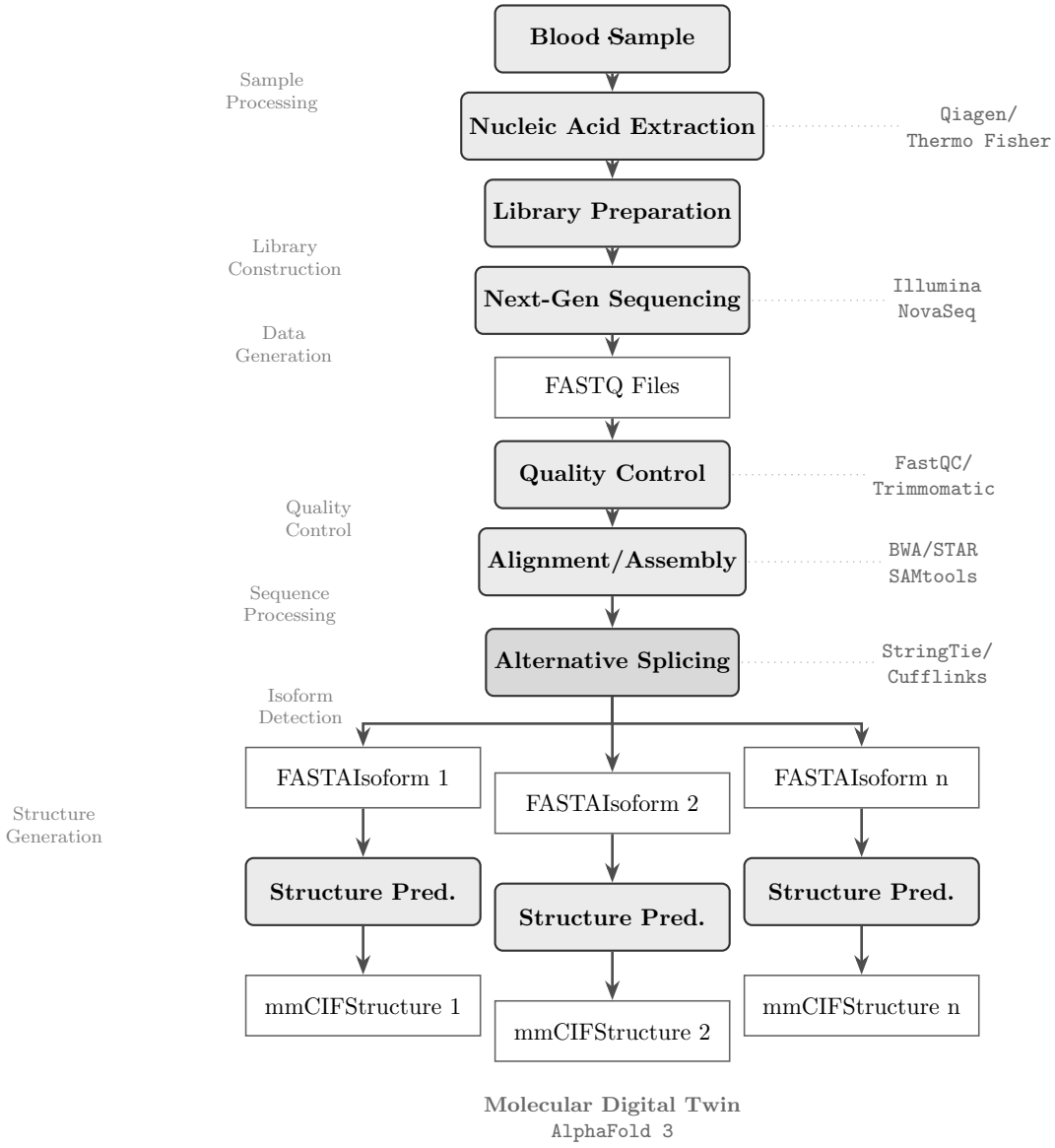

Figure 2: End-to-end workflow for nucleic acid identification and structural representation in blood digital twin construction. The pipeline transforms blood samples into DNA/RNA sequences and 3D structures through next-generation sequencing, quality control, sequence alignment, and AI-based structure prediction. Gray boxes indicate processing steps, white boxes represent data formats, and monospace text shows the primary computational tools employed at each stage.

6.10 End-to-End Workflow Summary

The complete pipeline from blood sample to nucleic acid structural representation is illustrated in Figure 2. This workflow integrates clinical-grade sequencing methods with AI-based structure prediction to generate molecular-level representations of blood DNA and RNA. The pipeline begins with biological sample collection and nucleic acid extraction, proceeds through library preparation and high-throughput sequencing, and culminates in sequence alignment, FASTA generation, and three-dimensional structural modeling. This workflow is validated in genomics facilities including the Broad Institute, TCGA, CPTAC [23], and FDA/CLIA-accredited laboratories.

7 Metabolite and Ligand Identification Workflow

Metabolites and small molecule ligands—including endogenous metabolites, cofactors, hormones, and pharmaceutical compounds—represent a critical component of blood molecular composition. These molecules participate in metabolic pathways, signal transduction, and therapeutic interventions. This section describes the complete workflow for converting blood samples to metabolite structures and protein-ligand complex models.

7.1 Blood Sample Preparation and Metabolite Extraction

Blood metabolomics begins with plasma or serum separation from whole blood through centrifugation. Plasma (anticoagulated blood) or serum (clotted blood) provides the acellular fraction containing soluble metabolites. Protein precipitation using organic solvents (methanol, acetonitrile) removes high-abundance proteins that interfere with mass spectrometry analysis. Metabolites are extracted using liquid-liquid or solid-phase extraction methods optimized for compound polarity ranges [4, 45]. Sample preparation protocols are standardized across clinical metabolomics facilities to ensure reproducibility and minimize batch effects.

7.2 Liquid Chromatography-Mass Spectrometry Analysis

Extracted metabolites undergo liquid chromatography-mass spectrometry (LC-MS/MS) analysis using high-resolution instruments such as Orbitrap, Q-Exactive, or Q-TOF systems [22]. Reversed-phase or hydrophilic interaction chromatography (HILIC) separates compounds based on chemical properties before mass spectrometric detection. Tandem mass spectrometry (MS/MS) generates fragmentation patterns diagnostic of molecular structure. Data-dependent acquisition (DDA) or data-independent acquisition (DIA) modes capture comprehensive spectral information. Instruments produce proprietary RAW files containing retention times, mass-to-charge ratios, and fragment ion spectra.

7.3 Conversion from RAW to mzML Format

RAW files are converted to the open mzML format using ProteoWizard's msconvert tool [8, 9], identical to the protein workflow (Section 5). This conversion ensures compatibility with metabolomics analysis software while preserving spectral data integrity and metadata. Peak-picking algorithms reduce file size by centroiding high-resolution profile data into discrete mass peaks.

7.4 Metabolite Identification and Database Matching

Metabolite identification matches experimental mass spectra against reference databases. The Human Metabolome Database (HMDB) [4] provides comprehensive coverage of endogenous human metabolites with associated spectral data, chemical structures, and biological annotations. METLIN [46] offers an extensive collection of metabolite tandem mass spectra. The Global Natural Products Social Molecular Networking (GNPS) [47] platform enables spectral library searching and molecular networking analysis. MassBank contributes additional high-quality reference spectra. Identification confidence is assessed through accurate mass matching (typically within 5 ppm), isotope pattern analysis, and MS/MS spectral similarity scoring. Identified compounds receive database accession identifiers (e.g., HMDB IDs, METLIN IDs).

7.5 Chemical Structure Generation

Database identifiers are used to retrieve chemical structures in standardized formats. Simplified Molecular Input Line Entry System (SMILES) notation provides a text-based representation of molecular structure encoding atom connectivity and stereochemistry. Structure Data File (SDF) and MOL2

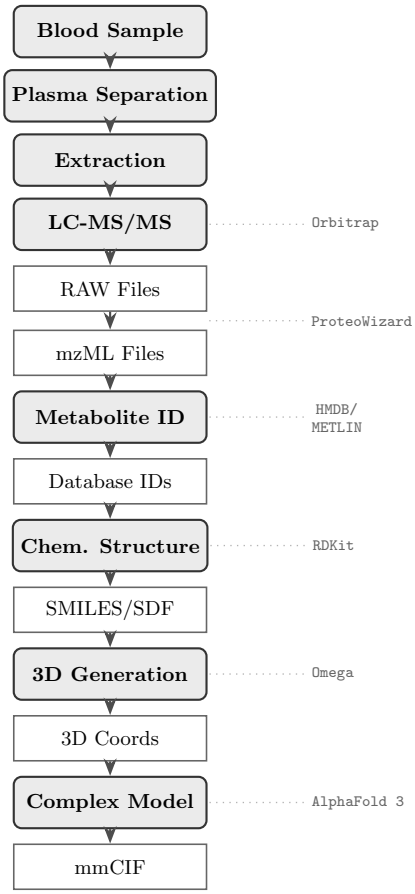

Figure 3: End-to-end workflow for metabolite and ligand identification with protein-ligand complex modeling. The pipeline transforms blood samples into 3D ligand structures and protein-ligand complexes through LC-MS/MS analysis, database matching, cheminformatics processing, and AI-based complex prediction. Gray boxes indicate processing steps, white boxes represent data formats, and monospace text shows the primary computational tools employed at each stage.

formats store two-dimensional chemical graphs with atomic coordinates and bond information. Cheminformatics tools such as RDKit [48] and OpenBabel [49] convert between chemical formats and validate structural integrity. These tools ensure consistency in chemical representation across different software platforms.

7.6 Three-Dimensional Ligand Structure Generation

Two-dimensional chemical structures are converted to three-dimensional conformations using computational chemistry methods. RDKit [48] employs distance geometry algorithms to generate initial 3D coordinates followed by force field optimization. OpenEye Omega [50] produces multiple low-energy conformers representing different spatial arrangements of flexible molecules. Conformer generation accounts for rotatable bonds, ring systems, and stereochemistry. Quality metrics include conformational energy, geometric strain, and Lipinski rule compliance for drug-like properties. Generated structures are stored in formats compatible with molecular modeling software (PDB, MOL2, SDF).

7.7 Protein-Ligand Complex Modeling

Three-dimensional ligand structures are combined with protein structures (from Section 5) for protein-ligand complex prediction. AlphaFold 3 [1] accepts protein sequences (FASTA format) and ligand structures (SMILES format) as input, specified in JSON format according to the AlphaFold 3 input schema. The prediction generates mmCIF files containing atomic coordinates for the entire protein-ligand complex with per-residue and per-atom confidence metrics. This capability enables computational prediction of drug binding sites, metabolite-protein interactions, and cofactor positioning within enzyme active sites. Predicted complexes inform drug discovery efforts and mechanistic understanding of metabolic pathways.

7.8 End-to-End Workflow Summary

The complete pipeline from blood sample to protein-ligand complex is illustrated in Figure 3. This workflow integrates clinical metabolomics with cheminformatics and AI-based structure prediction to generate molecular-level representations of blood metabolites and their interactions with proteins. The pipeline begins with sample preparation and metabolite extraction, proceeds through LC-MS/MS analysis and database matching, and culminates in chemical structure generation, 3D conformer generation, and protein-ligand complex modeling. This workflow is validated in metabolomics facilities including the NIH Metabolomics Workbench, Broad Institute, MD Anderson Cancer Center, and CPTAC metabolomics programs [23].

8 Cohesive Digital Twin of Human Blood

The preceding workflows (Sections 5–7) enable molecular-level characterization of blood proteins, nucleic acids, and metabolites. This section synthesizes these capabilities into a cohesive digital twin framework, clarifying which blood components are currently modelable at atomic resolution and which remain beyond current structural prediction capabilities. Figure 4 provides a visual overview of modelable versus non-modelable blood components.

8.1 Molecular-Level Modeling Capabilities

The majority of blood molecular components can be represented as three-dimensional structures suitable for computational analysis and drug discovery applications. Proteins identified through mass spectrometry (Section 5) are modeled using AlphaFold 3 [1], generating atomic coordinates for individual polypeptides and protein-protein complexes. DNA and RNA sequences from next-generation sequencing (Section 6) can be structurally modeled for nucleic acid structures and protein-nucleic acid complexes. Metabolites and small molecule ligands identified through LC-MS/MS (Section 7) are represented as three-dimensional conformers and integrated into protein-ligand complex predictions.

In addition to these major macromolecular classes, simpler blood constituents are readily modeled. Dissolved gases (O_2, CO_2, NO) and ions (Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, HCO_3^-) have well-characterized molecular geometries and coordination properties. Pathogens including viruses, bacteria, fungi, and parasites are identified through PCR-based methods, metagenomic sequencing, or immunoassays, with their nucleic acids processed through the sequencing pipeline (Section 6), proteins through proteomics (Section 5), and metabolites through metabolomics (Section 7). These elements can be incorporated into

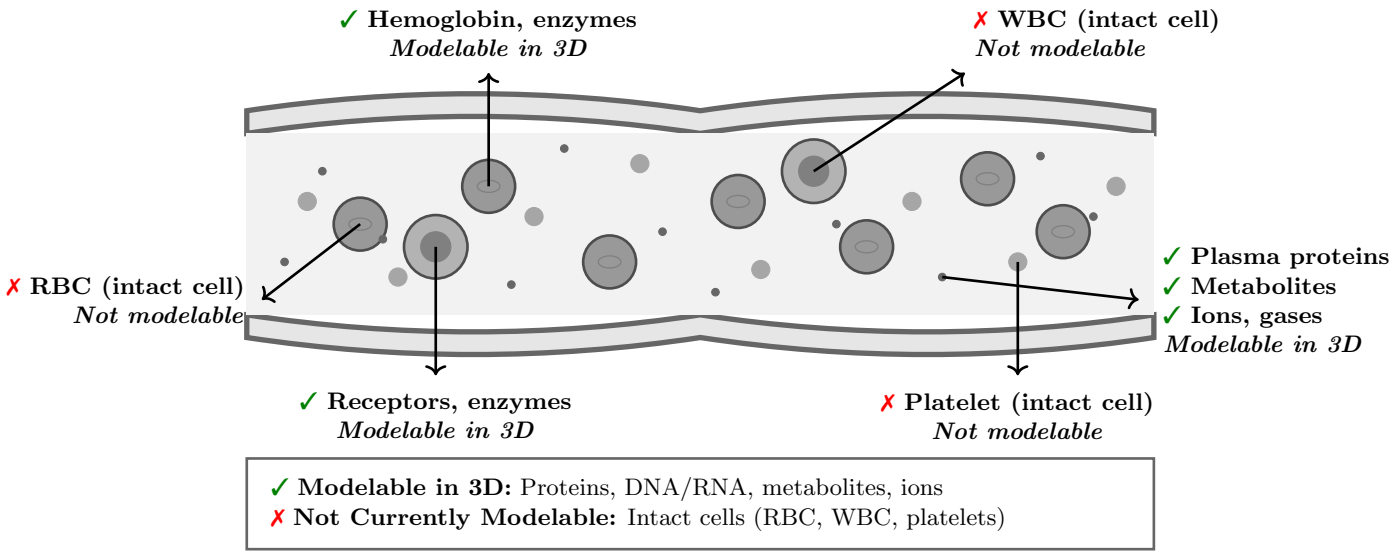

Figure 4: Conceptual representation of blood components and their current 3D modeling capabilities. Individual molecular components (proteins, nucleic acids, metabolites, ions) within cells and plasma are modelable at atomic resolution using AI-based structure prediction. However, intact cellular architectures (red blood cells, white blood cells, platelets) with their complete membrane topology, organelle organization, and spatial arrangement remain beyond current atomistic modeling approaches. Green checkmarks (✓) indicate components that can be structurally modeled in 3D; red crosses (✗) indicate components that cannot yet be modeled at cellular scale.

molecular assemblies, bundles, and complexes alongside proteins, nucleic acids, and metabolites to represent physiologically relevant molecular environments.

8.2 Current Limitations: Cellular Architecture

While molecular components are modelable, intact cellular structures present significant challenges. Red blood cells (erythrocytes), white blood cells (leukocytes), and platelets (thrombocytes) are multi-scale systems requiring representation of cellular membranes, organelles, cytoplasmic architecture, and spatial organization of thousands of molecular components. Modeling a complete cell requires not only structural information for individual proteins and nucleic acids but also accurate positioning of these molecules within cellular compartments, membrane topology, and dynamic interactions among components.

The internal architecture of cells—including nuclear envelopes, mitochondrial membranes, endoplasmic reticulum networks, and cytoskeletal frameworks—remains beyond the scope of current atomistic modeling approaches due to system size and computational complexity. This creates a gap between molecular-level digital twins (achievable today) and cellular-level digital twins (future capability). However, as hematology, cryo-electron microscopy, and computational biology advance, modeling complete cellular systems at molecular resolution becomes increasingly viable. Importantly, the foundational molecular components of cells—the proteins, nucleic acids, and metabolites that constitute cellular machinery—are currently modelable, creating immediate opportunities for computational analysis and therapeutic development.

8.3 Complexes, Assemblies, and Vesicles

Beyond individual molecular structures, blood contains numerous multi-component assemblies that can be modeled at atomic resolution. These complexes represent functional units critical to blood physiology and therapeutic targeting.

Plasma complexes include protein-protein assemblies such as complement cascade components, coagulation factor complexes (e.g., Factor VIIIa-Factor IXa, prothrombin activator complex), and immunoglobulin structures. Lipoprotein particles (LDL, HDL, VLDL) represent nano-scale assemblies of apolipoproteins, lipids, and cholesterol that can be modeled as protein-lipid complexes. Al-

phaFold 3's multi-chain prediction capability [1] enables structural modeling of these heteromeric assemblies, including apolipoprotein conformations and lipid-binding interfaces.

White blood cell surface complexes include T-cell receptor complexes, major histocompatibility complex (MHC) molecules bound to antigenic peptides, and cytokine receptor assemblies (e.g., interleukin receptors, interferon receptor complexes). Intracellular signaling protein complexes such as kinase cascades (MAPK pathway components, PI3K-AKT assemblies), ubiquitin ligase assemblies (E1-E2-E3 enzyme cascades), and transcription factor complexes are modelable from sequences identified through proteomic workflows. These complexes represent therapeutic targets for immunomodulatory drugs and biologics.

Red blood cell membrane complexes include hemoglobin tetramers (alpha-beta dimers), band 3 protein assemblies (anion exchanger complexes), spectrin-ankyrin networks, and glycophorin complexes that maintain erythrocyte shape and deformability. Cytoplasmic enzyme complexes involved in glycolysis (hexokinase, phosphofructokinase, pyruvate kinase) and pentose phosphate pathways (glucose-6-phosphate dehydrogenase complexes) are modelable from proteomic data.

Platelet complexes include integrin receptor assemblies (glycoprotein IIb/IIIa heterodimers), von Willebrand factor binding complexes, P-selectin adhesion molecules, and granule storage proteins (alpha-granule and dense-granule contents including platelet factor 4, beta-thromboglobulin, ADP, serotonin). These complexes are critical for hemostasis and represent targets for antiplatelet therapeutics such as abciximab and eptifibatide.

The ability to model these multi-component assemblies extends the digital twin framework beyond isolated molecules to functional biological units. This capability enables computational prediction of drug binding to complex assemblies, analysis of disease-associated structural perturbations in multi-protein interfaces, and rational design of therapeutics targeting protein-protein interaction surfaces. The integration of individual molecular structures into higher-order assemblies represents the transition from molecular catalogs to functional digital representations of blood biology.

9 Applications

This unified molecular representation of blood enables several transformative applications in biomedical research and clinical practice.

9.1 In Silico Clinical Trials

Once a comprehensive molecular digital twin of human blood is established, it becomes possible to conduct preliminary clinical trial simulations on digital representations rather than exclusively on human subjects. Traditional clinical trials follow multi-phase regulatory frameworks established by agencies such as the U.S. Food and Drug Administration (FDA) and equivalent international bodies, requiring extensive human testing across Phase I through Phase IV trials. A molecular digital twin framework offers the potential to supplement early-stage safety and efficacy assessments through computational modeling.

Initial simulations using blood digital twins will have limited predictive accuracy due to the complexity of biological systems and incomplete representation of physiological context. However, as databases of molecular blood twins expand through accumulation of diverse population samples, machine learning models trained on these datasets can improve predictive performance iteratively. This approach enables preliminary identification of adverse reactions, dosing parameters, and pharmacokinetic properties before proceeding to human trials, potentially reducing the time and cost associated with early-stage drug development while maintaining rigorous safety standards for human testing.

9.2 Accelerated Drug Discovery

Drug molecules are fundamentally molecular structures that interact with specific biological targets. By modeling interactions between candidate drug molecules and the molecular digital twin of human blood containing pathogens or disease markers, researchers can computationally identify compounds that bind effectively to target antigens or pathogenic elements. Drug molecules are characterized through the metabolomics and ligand identification pipeline (Section 7), enabling their structural representation and integration with protein targets from Section 5. This molecular-level simulation also reveals unintended interactions with other blood components, providing mechanistic insight into potential side effects observed in clinical use.

For a single pathogenic target, pharmaceutical databases may contain hundreds of thousands of candidate molecules. Traditional screening approaches require sequential in vitro and in vivo testing of each compound—a resource-intensive process spanning years. Molecular digital twins enable parallel in

silico screening of large compound libraries, identifying lead candidates that demonstrate optimal binding affinity, specificity, and minimal off-target effects. AlphaFold 3's protein-ligand complex prediction capability [1] facilitates rapid computational assessment of drug-target interactions before experimental validation. This computational pre-screening dramatically reduces the number of compounds requiring wet-lab validation, accelerating the drug discovery pipeline from target identification to clinical candidate selection.

9.3 Personalized Medicine and Precision Therapeutics

Human populations exhibit substantial genetic and epigenetic variation driven by evolutionary history, environmental exposures, and individual mutations. This biological diversity means that universal therapeutic approaches do not achieve uniform efficacy across all patients. Adverse drug reactions, including severe responses to vaccines or medications, often result from individual-specific molecular variations in drug metabolism, immune response, or target receptor structure.

Molecular digital twins of individual patients' blood enable personalized therapeutic strategies [28]. For blood-related conditions or diseases with systemic biomarkers, clinicians can computationally test candidate medications against a patient's specific molecular profile before administration. This approach identifies optimal therapeutic compounds, predicts individual-specific dosing requirements, and flags potential adverse reactions based on the patient's unique molecular landscape. As digital twin construction becomes more comprehensive—incorporating genomic, proteomic, metabolomic, and immunological data—personalized medicine transitions from reactive treatment adjustment to proactive, precision-guided therapeutic design.

9.4 Additional Applications

Beyond these primary use cases, molecular blood digital twins support systems biology simulations [27] that model complex physiological processes, disease progression dynamics, and multi-organ system interactions. Integration with pharmacokinetic and pharmacodynamic models enables prediction of drug distribution, metabolism, and clearance pathways. Digital twins also facilitate biomarker discovery through computational identification of disease-associated molecular signatures, support development of companion diagnostics, and enable longitudinal health monitoring through periodic molecular profiling. Molecular dynamics simulations [26] can further refine structural predictions and model protein-ligand interactions. As the framework matures, applications will extend to emerging areas including immunotherapy optimization, microbiome-host interactions, and precision nutrition.

10 Limitations

The molecular-level digital twin framework presented here captures blood components at atomic resolution but does not extend to complete cellular architectures. As discussed in Section 8.2, intact red blood cells, white blood cells, and platelets represent multi-scale systems requiring representation of cellular membranes, organelles, cytoplasmic organization, and spatial positioning of thousands of molecular components. Current atomistic modeling approaches cannot accommodate the system size and computational complexity required for complete cellular digital twins.

Additionally, predicted molecular structures remain approximations requiring experimental validation. AlphaFold 3 predictions provide confidence metrics (pLDDT scores) that guide reliability assessment, but experimentally determined structures from X-ray crystallography, cryo-electron microscopy, or NMR spectroscopy remain the gold standard for atomic-resolution structural information. Blood is a dynamic system with transient protein-protein interactions, metabolite fluctuations, and post-translational modifications that evolve over time. Static structural models capture molecular conformations but do not fully represent the temporal dynamics of blood biochemistry. The temporal resolution of current profiling methods may not capture rapid fluctuations in blood composition during acute physiological perturbations such as immune responses or coagulation cascades.

11 Future Work

Future developments include integration with molecular dynamics simulations to capture temporal evolution of molecular structures and protein-protein interaction dynamics. Time-resolved blood profiling using longitudinal sampling and multi-omics integration will enable construction of patient-specific digital blood twins for precision medicine applications, tracking molecular changes during disease progression and therapeutic intervention.

As computational capabilities advance and whole-cell modeling methodologies mature, the digital twin framework can be extended from molecular-level representations to complete cellular architectures. Advances in cryo-electron tomography, spatial transcriptomics, and super-resolution microscopy will provide spatial positioning constraints for molecular components within intact cells. Machine learning approaches integrating structural predictions with experimental spatial data may enable reconstruction of cellular organization at molecular resolution, bridging the current gap between molecular and cellular digital twins discussed in Section 8.2.

Advanced machine learning approaches may further refine structural predictions through integration of evolutionary constraints, experimental restraints, and physics-based energy functions. Real-time monitoring of blood composition through integration with wearable biosensors, microfluidic devices, and point-of-care diagnostic platforms will enable continuous updating of patient-specific digital twins, supporting dynamic clinical decision-making and early detection of disease onset or therapeutic response.

12 Conclusion

This framework presents a systematic approach to constructing a molecular-level digital twin of human blood through integration of mass spectrometry-based proteomics, next-generation sequencing, metabolomics, and AI-based structure prediction. The comprehensive workflows for proteins (Section 5), nucleic acids (Section 6), and metabolites (Section 7) enable identification and three-dimensional structural modeling of the majority of blood molecular components. The resulting digital twin (Section 8) encompasses individual molecular structures, multi-component assemblies, and functional complexes that can be computationally analyzed for drug discovery, disease mechanism elucidation, and personalized therapeutic development.

While intact cellular architectures remain beyond current modeling capabilities, the foundational molecular components—proteins, nucleic acids, metabolites, and their assemblies—are currently representable at atomic resolution. This molecular-level digital twin establishes the infrastructure for computational hematology, enabling in silico experimentation and hypothesis testing prior to costly and time-intensive wet-laboratory validation. As computational biology, structural biology, and clinical profiling technologies continue to advance, the framework presented here provides the foundation for increasingly comprehensive and personalized digital representations of human blood.

By bridging traditional hematology with modern computational structural biology, this approach enables unprecedented opportunities for systems-level understanding of blood physiology, rational therapeutic design, and precision medicine implementation. The molecular digital twin of human blood represents a transformative capability for biomedical research and clinical practice.

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